STA 431 MIDTERM II

Homework 4–5

with STAM2

Every problem, every subpart

App inputs, results, and written mathematics

A complete calculator walkthrough

All thirteen common/STA 431 questions and all three optional STA 631 extensions. Includes the new homework input loader, full marginal tables, joint-slice solver, and an explanation of the improvements made during testing.

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October 27, 2026

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How to use this walkthrough

This companion covers **every subpart of all thirteen common/STA 431 Homework 4–5 questions**, plus the three separately marked STA 631 extensions. Each problem has its app route, actual inputs, output targets, mathematical write-up, and an explanation of what the app saves you from calculating manually.

Use the updated ten-program package. Transfer all ten `.8xp` files together; the update adds `M2RUN` and `M2SLICE` and replaces the matching earlier solver files. Start with `PRGM > STAM2 > ENTER > ENTER`. The shortest route is

`HOMEWORK MAP > LOAD + SOLVE > homework group > question.`

The loader supplies the model's inputs to the general solver; it does not store final answers. For a problem with changed numbers, use the manual route in its section. Double-integral presets ask for an even resolution N; enter 4 for the polynomial homework models and compare the $N = 4$ and $N = 8$ outputs.

Homework group	Questions
`HW4 COMMON +431`	Q1 joint PMF; Q2 joint PDF; Q3 device lifetime; Q4 PMF events; Q5 $X < Y$.
`HW5 Q1 TO Q4`	PMF moments; continuous square; exponential MGF; affine MGF.
`HW5 Q5 TO Q8`	Insurance variance; three-variable variance; conditional moments; joint moments.
`OPTIONAL 631`	HW4 triangular density and reimbursement; HW5 zero covariance with dependence.

Reading calculator displays. Press `ENTER` to advance through pages. Use `BACK` to return through menus. The actual $\leq$ comparison is one token under `[2nd][MATH] > TEST`; enter negative numbers with the unary `(-)` key. In displayed formulas and input fields, uppercase calculator X, Y represent the lowercase mathematical variables x, y. In MGF tools, calculator X represents the transform argument t.

What to submit as mathematical work. Copy the variable definition or event, support, relevant sum/integral or moment formula, and the exact conclusion. Numerical integrals and suggested fractions support a calculation; the short derivations below establish exact forms and convergence conditions. A marginal density such as $f_Y(1/2)$ is a density value, not the probability $P(Y = 1/2)$.

Meaning of the double-integral result. `RAW I` is the integral of the supplied q over the entered region. For an already normalized joint PDF restricted to an event, it is the event probability. `NORMALIZE FULL S` applies to the full support of a model you intend to normalize. Do not renormalize the survival or $X < Y$ event regions to one.

Verification. The actual tokenized programs were exercised through all sixteen homework presets, the new slice calculations, marginal tables, and launcher returns. These are simulator checks; physical CE execution remains to be checked using the package's device examples.

1 Homework 4: every problem with STAM2

This walkthrough covers the three common questions, both STA 431 questions, and the two separately labeled STA 631 questions. Start with `PRGM > STAM2`. The routes below name the actual solver menus. For the first five questions, use `HOMEWORK MAP > LOAD + SOLVE`, then `HW4 COMMON +431` and the matching question. This preloads inputs, not answers: the general solvers still calculate each result. Manual entries below show the model being solved and let you handle altered exam numbers.

For continuous calculations, a displayed fraction is a numerical result that suggests an exact value. The integral and its algebra establish the exact answer. In particular, the app does not symbolically derive a marginal-density formula or prove independence of continuous variables.

1.1 Common Question 1: joint PMF, marginals, conditioning, independence

The joint PMF is $p(x, y) = c(x + y^2)$ on $x \in \{1, 2\}$ and $y \in \{1, 2, 3\}$.

Quick load: choose `Q1 JOINT PMF`; continue past its model screen.

Manual alternative: load the model once. Choose `DISCRETE + JOINT > FORMULA/GRID`:

Prompt	Enter
`x VALUES LIST?`	`{1,2}`
`y VALUES LIST?`	`{1,2,3}`
`q(X,Y)?`	`X+Y^2`

The app constructs all six pairs, sums their weights to 37, and displays $c = 1/37$. You do not have to enter c or manually type the six joint probabilities.

(a) What to write. The six weights are $2, 5, 10, 3, 6, 11$, hence $1 = c(37)$ and $\boxed{c = 1/37}$.

(b) Marginals. From `TABLE TOOLS`, choose `MARG/CONDITIONAL`. Choose the new `MARGINAL TABLES` option to display both complete marginal tables. Press `ENTER` between them. You no longer have to query the five values individually. The returned values are

x	1	2
$p_X(x)$	17/37	20/37

y	1	2	3
$p_Y(y)$	5/37	11/37	21/37.

For the general formulas, write

$$p_X(x) = \sum_{y=1}^{3} \frac{x + y^2}{37} = \frac{3x + 14}{37}, \quad x = 1, 2;$$

$$p_Y(y) = \sum_{x=1}^{2} \frac{x + y^2}{37} = \frac{3 + 2y^2}{37}, \quad y = 1, 2, 3.$$

Both PMFs are zero outside their supports.

(c) Conditional PMF. Choose `X GIVEN Y` and enter 2. The conditional-distribution screen gives

$$\boxed{p_{X|Y}(1 \mid 2) = 5/11, \qquad p_{X|Y}(2 \mid 2) = 6/11.}$$

The written step is $(5/37)/(11/37) = 5/11$ and $(6/37)/(11/37) = 6/11$. These sum to one. Extra conditional moments shown afterward are not required by this question.

(d) Independence. Choose `INDEPENDENCE`. The app reports a numerical factorization failure and now displays a violating pair with its joint probability and marginal product. Convert that observation into an exact comparison. For example:

$$p(1,1) = \frac{2}{37} = \frac{74}{1369} \neq \frac{17}{37}\frac{5}{37} = \frac{85}{1369} = p_X(1)p_Y(1).$$

Therefore X, Y are dependent. One failing PMF pair is sufficient.

Why this is easier. The homework loader removes model re-entry; the manual alternative needs just three model entries. The new marginal-table command displays all five marginal values together, and the independence tool supplies a counterexample pair. The same data stay loaded for all subparts. You still write the marginal sums and the exact independence comparison.

1.2 Common Question 2: a joint PDF and conditional-density formula

Here $f(x, y) = c(2x + y)$ on $0 < x < 1$, $0 < y < 2$.

Quick load: choose `Q2 JOINT PDF`, then enter outer resolution $N = 4$.

(a) Manual normalization route. Choose `DENSITY + INTEGRAL > DOUBLE INTEGRAL`. Enter the following model. The fixed parameter A is unused, so enter 0.

Prompt	Enter
`q(X,Y)?`	`2X+Y`
`FIXED A?`	`0`
`LOWER X?, UPPER X?`	`0, 1`
`LOWER Y(X)?, UPPER Y(X)?`	`0, 2`
`g(X,Y),1=AREA?`	`1`
`EVEN N 4-100?`	`4`

The app displays raw integral $I \approx 4$ at $N = 4$ and $N = 8$. Choose `NORMALIZE FULL S`; because the entered rectangle is the full support, the app gives $c \approx 1/4$.

What to write for (a).

$$1 = c \int_0^1 \int_0^2 (2x + y)\, \mathrm{d}y\, \mathrm{d}x = c \int_0^1 (4x + 2)\, \mathrm{d}x = 4c, \qquad \boxed{c = \frac{1}{4}}.$$

The numerical solver removes the arithmetic burden, while this short calculation establishes the exact constant.

(b) Derive the marginal of Y. Hold y fixed and integrate over x:

$$\boxed{f_Y(y) = \int_0^1 \frac{2x + y}{4}\, \mathrm{d}x = \frac{1 + y}{4}, \quad 0 < y < 2;}$$

$f_Y(y) = 0$ otherwise. The app can check values of this function, but it cannot return the symbolic expression for all y.

Numerical check of (b), using the improved joint-slice tool. Choose `DENSITY + INTEGRAL > JOINT SLICE > X GIVEN Y`. Enter joint shape `2X+Y`, known $c = 1/4$, given value 1, and remaining-variable bounds 0 and 1. The results screen reports marginal density $f_Y(1) \approx 1/2$. To check another y, use `NEW GIVEN VALUE`; the same joint model remains loaded. This separates the global multiplier from the conditional normalization instead of making you rename variables.

(c) Derive the conditional density.

$$\boxed{f_{X|Y}(x \mid y) = \frac{f(x, y)}{f_Y(y)} = \frac{2x + y}{1 + y}, \quad 0 < x < 1, \quad 0 < y < 2.}$$

For each allowed y, it is zero outside $0 < x < 1$. The denominator belongs to the given variable Y.

In the joint-slice tool, choose `PDF AT POINT` and enter remaining value $x = 1/2$. It reports conditional density approximately 1 and joint density approximately $1/2$, agreeing with $(2x + 1)/2$ and $(2x + 1)/4$. Choose `INTERVAL PROB` with event bounds 0 and $1/2$ to obtain approximately $3/8$. These numerical checks are optional; the homework's requested density is the function boxed above.

Why this is easier. The app handles normalization and conditional arithmetic. You still derive the requested symbolic functions from the short integrals above.

1.3 Common Question 3: a device that fails when either component fails

The joint PDF is $(x+y)/8$ on the square $(0,2)^2$. The device lifetime is $T = \min(X, Y)$, so the requested event is

$$\{T \le 1\} = \{X \le 1\} \cup \{Y \le 1\}.$$

Its complement is the single rectangle $X > 1, Y > 1$.

Quick load: choose `Q3 SERIES LIFETIME`, continue past the survival/complement reminder, and enter $N = 4$.

Manual route. Choose `DENSITY + INTEGRAL > DOUBLE INTEGRAL`:

Input	Value
Joint integrand `q(X,Y)`	`(X+Y)/8`
Fixed parameter `A`	`0`
Outer x limits	`1, 2`
Inner y limits	`1, 2`
Multiplier `g(X,Y)`	`1`
Outer resolution `N`	`4`

Read `RAW I`, approximately 3/8, on both resolution screens. This is the survival probability because you entered the already normalized joint density over the survival rectangle. Do *not* choose `NORMALIZE FULL S`: the entered rectangle is an event region, not the original support.

Subtract the survival probability from one:

$$\boxed{P(T \le 1) = 1 - \frac{3}{8} = \frac{5}{8} = 0.625.}$$

You can do $1 - 3/8$ on the home screen after leaving the solver. The preset also reminds you that failure is `1-RAW I`; the solver itself computes the survival integral.

What to write.

$$P(T \le 1) = 1 - P(X > 1, Y > 1) = 1 - \int_1^2 \int_1^2 \frac{x+y}{8}\, dy\, dx = \frac{5}{8}.$$

The event translation earns the reasoning; the app handles the integration and arithmetic. Do not multiply marginal survival probabilities, because independence was not given.

Why this is easier. A complement turns an L-shaped union into one rectangle. The app then computes its double integral in one run, with no separate intersection correction and no assumed independence.

1.4 STA 431 Question 4: two discrete event probabilities

The PMF is $p(x, y) = (x + y)/32$ for $x \in \{1, 2\}$ and $y \in \{1, 2, 3, 4\}$.

Quick load: choose `Q4 PMF EVENTS`; both event queries use the same loaded table.

Manual alternative: load once. Choose `DISCRETE + JOINT > FORMULA/GRID`. Enter x list `{1,2}`, y list `{1,2,3,4}`, and weight formula `X+Y`. The weight total is 32, so the app reconstructs the stated PMF automatically.

(a) $P(X > Y)$**.** From `TABLE TOOLS`, choose `EVENT PROB > P(EVENT)`. At `EVENT X,Y?`, enter `X>Y`. The app returns

$$\boxed{P(X > Y) = \frac{3}{32} = 0.09375.}$$

For the written solution, only the pair $(2, 1)$ satisfies the event, giving $p(2, 1) = (2 + 1)/32$.

(b) $P(X + Y \leq 3)$**.** You return to `EVENT TYPE`, so choose `P(EVENT)` again and enter `X+Y<=3`, using the actual $\leq$ token rather than separate `<` and `=` characters. The result is

$$\boxed{P(X + Y \leq 3) = \frac{1}{4} = 0.25.}$$

The contributing pairs are $(1, 1)$, $(1, 2)$, and $(2, 1)$, and

$$P(X + Y \leq 3) = \frac{2 + 3 + 3}{32} = \frac{8}{32} = \frac{1}{4}.$$

Why this is easier. You enter the support and weight formula once, then change only the event expression. The app selects the right pairs and adds their unequal weights. Counting qualifying pairs and dividing by eight would be wrong because the pairs are not equally likely.

A useful way to check your entry. The support is discrete. In part (a), Y cannot be 1.5; only the listed integer pairs matter. In part (b), the boundary pairs with sum exactly 3 must be included.

1.5 STA 431 Question 5: $P(X < Y)$ from a joint PDF

Use $f(x, y) = (2x + y)/4$ on $0 < x < 1$, $0 < y < 2$.

Build the event region. For a fixed $x \in (0, 1)$, the event $x < y$ and the support $y < 2$ give $x < y < 2$. Thus

$$P(X < Y) = \int_0^1 \int_x^2 \frac{2x + y}{4} \, \mathrm{d}y \, \mathrm{d}x.$$

The lower inner limit is x, not 0. Once that support intersection is correct, the calculator can do the entire numerical integration.

Quick load: choose `Q5 X LESS Y`, then enter $N = 4$.

Manual route. Choose `DENSITY + INTEGRAL > DOUBLE INTEGRAL`:

Input	Value
`q(X,Y)?`	`(2X+Y)/4`
`FIXED A?`	`0`
Outer x limits	`0, 1`
Inner y limits	`X, 2`
`g(X,Y),1=AREA?`	`1`
`EVEN N 4-100?`	`4`

Both resolution screens give `RAW I` approximately $19/24 = 0.7916666667$. Use this raw value as the probability. Normalizing the event region would answer a different question.

Exact written calculation.

$$\begin{aligned} P(X < Y) &= \int_0^1 \left[x + \frac{1}{2} - \frac{5}{8}x^2 \right] \mathrm{d}x \\ &= \frac{1}{2} + \frac{1}{2} - \frac{5}{24} = \boxed{\frac{19}{24}}. \end{aligned}$$

The app removes both numerical integrations; writing the first iterated integral shows that you understood the event. The simplified single integral establishes the exact fraction.

Why the chosen order is easier. Integrating first in y needs one region. Reversing the order forces a split at $y = 1$:

$$\int_0^1 \int_0^y f(x, y) \, \mathrm{d}x \, \mathrm{d}y + \int_1^2 \int_0^1 f(x, y) \, \mathrm{d}x \, \mathrm{d}y.$$

Both are correct, but the first setup requires one app run and fewer boundary decisions.

1.6 Optional STA 631 Question 4: triangular support

Here $f(x,y) = 24x(1-x-y)$ on $x>0, y>0, x+y<1$. This is extra practice; it is separate from the STA 431 Question 4.

Quick load: from `LOAD + SOLVE > OPTIONAL 631`, choose `HW4 Q4 TRIANGLE` and enter $N = 4$.

Manual normalization check. Use `DOUBLE INTEGRAL` with `q=24X*(1-X-Y)`, $A = 0$, outer x limits $0, 1$, inner y limits 0, `1-X`, multiplier 1, and $N = 4$. The raw integral is approximately 1, confirming the proposed normalization numerically. The sloping upper bound is the key entry.

(a) Marginal formulas require the appropriate slices.

$$\boxed{f_X(x) = \int_0^{1-x} 24x(1-x-y)\,dy = 12x(1-x)^2, \quad 0<x<1,}$$

$$\boxed{f_Y(y) = \int_0^{1-y} 24x(1-x-y)\,dx = 4(1-y)^3, \quad 0<y<1.}$$

Both densities are zero otherwise. Each upper limit comes from $x+y<1$.

Numerical checks of the two marginals, with original variable names. Choose `DENSITY + INTEGRAL > JOINT SLICE`. Use joint shape `24X*(1-X-Y)` and known $c = 1$ in both directions:

Target	Direction, fixed value, remaining-variable bounds	Marginal output
$f_X(1/2)$	`Y GIVEN X`; given $1/2$; bounds 0, `1-X`	$3/2$
$f_Y(1/2)$	`X GIVEN Y`; given $1/2$; bounds 0, `1-Y`	$1/2$

The first result exceeds one because it is a density value, not an interval probability. The app keeps the fixed coordinate in place while integrating the remaining one. These spot checks verify arithmetic at chosen values; the integral formulas above supply the complete marginal densities.

(b) Independence. The marginals are positive throughout $(0,1)$, but the joint PDF is zero wherever $x+y>1$. For example, on the positive-area rectangle $0.6<x<0.8$, $0.6<y<0.8$, the marginal product is positive while the joint PDF is zero. Therefore $f_{X,Y} \neq f_X f_Y$ on a set of positive area and the variables are dependent.

Why the app helps. It checks the normalization and evaluates marginal slices with a moving bound. The app cannot replace the symbolic marginal derivations or the support-based independence argument.

1.7 Optional STA 631 Question 5: conditional reimbursement

The joint density is $f(x, y) = (20 - x)/(25x)$ on $10 < x < 20$, $x/2 < y < x$. Here X is spending and Y is reimbursement.

(a) Marginal of spending. The density is constant in y over an interval of length $x/2$, so

$$\boxed{f_X(x) = \int_{x/2}^{x} \frac{20 - x}{25x}\, \mathrm{d}y = \frac{20 - x}{50}, \quad 10 < x < 20,}$$

with zero elsewhere. The interval length is more efficient than doing a long integration.

App check of (a). Choose `DENSITY + INTEGRAL > JOINT SLICE > Y GIVEN X`. Enter `q=(20-X)/(25X)`, known $c = 1$, given spending $X = 12$, lower reimbursement bound `X/2`, and upper bound `X`. The app reports marginal density $f_X(12) \approx 4/25$. Original X remains the spending variable; the app integrates in Y. It also reports conditional mean 9 and variance 3, which are useful checks but are not required here.

(b) Conditional density of reimbursement.

$$\boxed{f_{Y|X}(y \mid x) = \frac{(20 - x)/(25x)}{(20 - x)/50} = \frac{2}{x}, \quad \frac{x}{2} < y < x, \quad 10 < x < 20.}$$

Thus $Y \mid X = x$ is uniform on $(x/2, x)$. At $x = 12$ its density is $1/6$ on $(6, 12)$.

(c) Continue in the same joint-slice model. Choose `INTERVAL PROB`, enter event lower 8 and event upper 12, and read approximately

$$\boxed{P(Y \geq 8 \mid X = 12) = \frac{2}{3}.}$$

No density re-entry or variable renaming is needed. Alternatively, the quick-load route `OPTIONAL 631 > HW4 Q5 REIMBURSE` preloads the equivalent uniform slice on $(6, 12)$; choose its `INTERVAL PROB` and enter 8,12. Write the exact calculation as

$$P(Y \geq 8 \mid X = 12) = \int_{8}^{12} \frac{1}{6}\, \mathrm{d}y = \frac{12 - 8}{12 - 6} = \frac{2}{3}.$$

Why this is easier. Recognizing the uniform conditional law reduces the problem to an interval-length ratio. The app automatically divides the event integral by the full conditional-slice mass. You never divide by $P(X = 12)$, which is zero in this continuous model.

2 Homework 5: solve every problem with STAM2

These instructions cover Common Problems 1–7, the STA 431 Problem 8, and the separately marked STA 631 Problem 8. Start with `PRGM > STAM2`. The paths below begin at its main menu. Press `ENTER` to advance through result pages and use `BACK` to return through menus. For the shortest route, use `HOMEWORK MAP > LOAD + SOLVE` and select the matching question. The loader supplies inputs to the same general solvers; it does not merely display an answer. The manual routes below let you reproduce each calculation with changed data.

Use the calculator's actual X, Y, $e^{(}$, square, and comparison tokens. Enter a negative number with the unary `(-)` key. Fractions such as `1/4` can be entered directly. Fraction-looking outputs from numerical integration are convenient approximations; the displayed integral and its algebra establish the exact result.

2.1 Problem 1: normalize a PMF, then find its mean and variance

Quick load: group `HW5 Q1 TO Q4`, then `Q1 PMF MOMENTS`. Choose `MOMENTS` after normalization.

The model is $p_X(x) = c(x^2 + x)$ for $x = 1, 2, 3, 4$.

App route. Select `DISCRETE + JOINT > FORMULA/GRID`.

Prompt	Enter
`x VALUES LIST?`	`{1,2,3,4}`
`y VALUES LIST?`	`{0}`; this dummy coordinate makes a one-variable grid.
`q(X,Y)?`	`X^2+X`

The program generates the four weights $2, 6, 12, 20$, reports total weight 40, and displays $c = 1/40$. Choose `MOMENTS`. The X result page gives

$$\boxed{\mathbb{E}X = \frac{13}{4}, \qquad \mathbb{E}X^2 = \frac{227}{20}, \qquad \operatorname{Var} X = \frac{63}{80}.}$$

The app now recognizes the one-variable grid and suppresses dummy Y and correlation pages.

Write for parts (a)–(c).

$$1 = c(2 + 6 + 12 + 20) = 40c \quad \Rightarrow \quad c = \frac{1}{40},$$

$$\mathbb{E}X = \frac{2 + 12 + 36 + 80}{40} = \frac{13}{4}, \qquad \mathbb{E}X^2 = \frac{2 + 24 + 108 + 320}{40} = \frac{227}{20},$$

$$\operatorname{Var} X = \mathbb{E}X^2 - (\mathbb{E}X)^2 = \frac{227}{20} - \frac{169}{16} = \frac{63}{80}.$$

Why the app helps. One formula generates the values, sums the weights, normalizes them, and computes both moments and the variance. You do not need to enter four separate probabilities or repeat the weighted arithmetic.

2.2 Problem 2: a symmetric continuous variable and its square

Quick load: group `HW5 Q1 TO Q4`, then `Q2 PDF AND SQUARE`. Choose `MEAN+VARIANCE`, then `FUNCTION g(X)` with `X^2`.

The given density is $f_X(x) = \frac{3}{4}(1 - x^2)$ on $(-1, 1)$. Both X and $Y = X^2$ are involved.

App route. Select `DENSITY + INTEGRAL > 1D PDF OR SLICE`.

Prompt	Enter
`q(X)?`	`1-X^2`
`FIXED A?`	`0`; this model has no parameter A.
`LOWER X?`, `UPPER X?`	`(-)1`, then `1`

This manual route omits the known factor 3/4 because the tool normalizes its shape. It reports $\int q = 4/3$ and normalizing factor 3/4. A preset that retains the given normalized density instead reports total mass 1; both routes yield the same moments.

Choose `MEAN+VARIANCE`. For part (a), read

$$\boxed{\mathbb{E}X = 0, \qquad \mathbb{E}X^2 = \frac{1}{5}, \qquad \operatorname{Var} X = \frac{1}{5}.}$$

Then choose `FUNCTION g(X)` and enter `X^2`. The output labels now refer to $g(X) = Y$, so part (b) is

$$\boxed{\mathbb{E}Y = \frac{1}{5}, \qquad \mathbb{E}Y^2 = \frac{3}{35}, \qquad \operatorname{Var} Y = \frac{8}{175}.}$$

The program computes the centered integral for the transformed variance; you do not have to derive the distribution of X^2.

Write the mathematical setup. Symmetry makes $xf_X(x)$ odd, hence $\mathbb{E}X = 0$. For the even moments,

$$\mathbb{E}X^2 = \frac{3}{2}\int_0^1 (x^2 - x^4)\,\mathrm{d}x = \frac{1}{5}, \qquad \mathbb{E}X^4 = \frac{3}{2}\int_0^1 (x^4 - x^6)\,\mathrm{d}x = \frac{3}{35}.$$

Then $\mathbb{E}Y = \mathbb{E}X^2$ and $\operatorname{Var} Y = \mathbb{E}X^4 - (\mathbb{E}X^2)^2 = 3/35 - 1/25 = 8/175$.

Why the app helps. Enter the density once, then change only the transformation to X^2. The same solver performs all the required finite integrals and uses the fourth moment automatically when calculating $\operatorname{Var}(X^2)$.

2.3 Problem 3: derive an exponential MGF and obtain two moments

Quick load: group `HW5 Q1 TO Q4`, then `Q3 EXPONENTIAL MGF`. Advance through the moments and MGF pages.

Here $f_X(x) = 3e^{-3x}$ for $x > 0$, so the exponential *rate* is 3.

App route. Select `MGF TOOLS > EXPONENTIAL RATE`.

Prompt	Enter
`RATE L>0?`	`3`
`SCALE a?`	`1`
`SHIFT b?`	`0`

Scale 1 and shift 0 request the original variable. Advance through the output pages. They give

$$\boxed{M_X(t) = \frac{3}{3 - t}, \quad t < 3; \qquad \mathbb{E}X = \frac{1}{3}, \quad \mathbb{E}X^2 = \frac{2}{9}.}$$

The tool also reports variance 1/9. Its general transformed-MGF display simplifies to the original expression when $a = 1, b = 0$.

Write part (a), including convergence.

$$M_X(t) = \mathbb{E}(e^{tX}) = \int_0^\infty 3e^{-(3-t)x}\,\mathrm{d}x = \frac{3}{3-t} \quad \text{for } t < 3.$$

The improper integral converges precisely when $3 - t > 0$. A finite numerical cutoff would not establish this domain; the exponential template records the relevant analytic result.

Write part (b).

$$M_X'(t) = \frac{3}{(3-t)^2}, \qquad M_X''(t) = \frac{6}{(3-t)^3},$$

so $M_X'(0) = 1/3$ and $M_X''(0) = 2/9$.

Why the app helps. The exact exponential template supplies the MGF form, its domain, and both moment values from one parameter. Your written work then shows the short integral and derivative argument required by the question.

2.4 Problem 4: center and scale a given MGF

Quick load: group `HW5 Q1 TO Q4`, then `Q4 AFFINE MGF`. The original MGF and transformation are already supplied.

The MGF is $M_X(t) = e^{4t+8t^2}$ and $Z = (X-4)/4 = \frac{1}{4}X - 1$.

App route. Select `MGF TOOLS > QUADRATIC LOG MGF`.

Prompt	Enter
`C?`	`4`; coefficient of t in $\log M_X(t)$.
`D` ≥ 0`?`	`8`; coefficient of t^2, equal to half the original variance.
`SCALE a?`	`1/4`
`SHIFT b?`	`(-)1`

The original moments page gives mean 4, second moment 32, and variance 16. The transformed page gives mean 0, second moment 1, and variance 1. On `TRANSFORMED MGF`, the form is e^{Ft+Ht^2} with $F = 0$ and $H = 1/2$:

$$\boxed{M_Z(t) = e^{t^2/2}, \quad \mathbb{E}Z = 0, \quad \operatorname{Var} Z = 1.}$$

Both MGFs are finite for every real t.

Write part (a). Use the affine rule before simplifying:

$$M_Z(t) = e^{-t}M_X(t/4) = e^{-t}\exp\left(4\frac{t}{4} + 8\frac{t^2}{16}\right) = e^{t^2/2}.$$

For part (b), $M_Z'(t) = te^{t^2/2}$ and $M_Z''(t) = (1+t^2)e^{t^2/2}$, so $\mathbb{E}Z = 0$, $\mathbb{E}Z^2 = 1$, and $\operatorname{Var} Z = 1$.

Why the app helps. It keeps track of the shift, the substitution into the old MGF, and the factor of two between a quadratic log-MGF coefficient and a variance. This problem matches an exact template, so numerical differentiation is unnecessary.

2.5 Problem 5: variance of the updated insurance benefit

Quick load: group `HW5 Q5 TO Q8`, then `Q5 CLAIM VARIANCE`. Advance to the computed variance.

First express the new total as

$$T = W + V = 2X + 1.1Y + 100.$$

Only variances and covariance are supplied; a mean is not needed.

App route. Select `LINEAR COMBOS > ONE COMBINATION > VAR/COV ONLY`. Enter values in the order shown:

Prompt	Enter
`n (1-3)?`	`2`
Variance of $X_1 = X$, then $X_2 = Y$	`300`, then `500`
`COV(X1,X2)`	`150`
`a FOR U?` for X_1, then X_2	`2`, then `1.1` (or `11/10`)
`OFFSET c FOR U?`	`100`

The result is $\boxed{\mathrm{Var}(T) = 2465}$. The display correctly says `MEAN NOT PROVIDED`; do not fill in an invented mean.

Write the variance expansion.

$$\begin{aligned}\mathrm{Var}(T) &= 4\,\mathrm{Var}\,X + (1.1)^2\,\mathrm{Var}\,Y + 2(2)(1.1)\,\mathrm{Cov}(X,Y)\\ &= 4(300) + 1.21(500) + 4.4(150)\\ &= 1200 + 605 + 660 = 2465.\end{aligned}$$

The fixed increase of 100 contributes no variance.

Why the app helps. Enter the coefficient and covariance data once; it forms all square and cross terms. The variance-only mode avoids asking for information the problem does not supply.

2.6 Problem 6: a signed combination of three variables

Quick load: group `HW5 Q5 TO Q8`, then `Q6 THREE VARIABLES`. All supplied variances, covariances, and coefficients are prefilled.

The target is $W = 3X - 2Y + Z$.

App route. Select `LINEAR COMBOS > ONE COMBINATION > VAR/COV ONLY`.

Input group	Enter in this order
Number of variables	`3`
Variances of $X_1 = X$, $X_2 = Y$, $X_3 = Z$	`4`, `9`, `2`
Covariances 12, 13, 23	`1`, `(-)1`, `0`
Coefficients for X_1, X_2, X_3	`3`, `(-)2`, `1`
Offset	`0`

The result page gives $\boxed{\mathrm{Var}(W) = 56}$.

Write all three covariance terms.

$$\begin{aligned}\operatorname{Var}(W) &= 9\operatorname{Var}X + 4\operatorname{Var}Y + \operatorname{Var}Z \\ &\quad - 12\operatorname{Cov}(X,Y) + 6\operatorname{Cov}(X,Z) - 4\operatorname{Cov}(Y,Z) \\ &= 36 + 36 + 2 - 12 - 6 - 0 = 56.\end{aligned}$$

The coefficient -2 changes two cross-term signs. The negative covariance $\operatorname{Cov}(X,Z) = -1$ then makes $6\operatorname{Cov}(X,Z) = -6$.

Why the app helps. Three variances and three distinct covariances determine the calculation. It assembles every pair once with the correct signed coefficient products, reducing the chance of omitting a term or its factor of two.

2.7 Problem 7: marginal and conditional densities, then conditional moments

Quick load: group `HW5 Q5 TO Q8`, then `Q7 CONDITIONAL`. This loads the slice at $Y = 1/2$; select `MEAN+VARIANCE`.

The joint PDF is $f(x,y) = x + y$ on the unit square.

Parts (a)–(b): write the short symbolic calculation.

$$f_Y(y) = \int_0^1 (x+y)\,\mathrm{d}x = \frac{1}{2} + y, \qquad 0 < y < 1,$$

with zero density elsewhere. Therefore

$$f_{X|Y}(x \mid y) = \frac{x+y}{1/2+y}, \qquad 0 < x < 1, \quad 0 < y < 1,$$

and the conditional density is zero outside its x slice. The new `JOINT SLICE` tool now checks marginal and conditional density values directly. Select `DENSITY + INTEGRAL > JOINT SLICE > X GIVEN Y`, enter `q(X,Y)=X+Y`, known multiplier `1`, given value `1/2`, and remaining-variable bounds `0`, `1`. It reports the marginal density $f_Y(1/2) = 1$ alongside the conditional moments below. In `SLICE OPTIONS`, `NEW GIVEN VALUE` lets you change y without retyping the model; `PDF AT POINT` evaluates the conditional and joint densities at a remaining x value. For example, at $y = 1/4$ the marginal is $3/4$, and at $x = 1/4$ the conditional density is $2/3$. These are density values, not probabilities at isolated points. A numerical calculation verifies values of these functions; it does not produce a symbolic formula in y.

Part (c): app route. Select `DENSITY + INTEGRAL > 1D PDF OR SLICE`.

Prompt	Enter
`q(X)?`	`X+A`
`FIXED A?`	`1/2`; A is the given value of Y.
`LOWER X?`, `UPPER X?`	`0`, then `1`

The slice mass is $D = f_Y(1/2) = 1$. Select `MEAN+VARIANCE`. Read

$$\boxed{\mathbb{E}(X \mid Y = 1/2) = \frac{7}{12}, \qquad \mathbb{E}(X^2 \mid Y = 1/2) = \frac{5}{12}, \qquad \operatorname{Var}(X \mid Y = 1/2) = \frac{11}{144}.}$$

For your written calculation,

$$\int_0^1 x(x+1/2)\,\mathrm{d}x = \frac{7}{12}, \qquad \int_0^1 x^2(x+1/2)\,\mathrm{d}x = \frac{5}{12},$$

then $5/12 - (7/12)^2 = 11/144$.

Why the app helps. Enter the unnormalized joint slice $X + A$ directly. The solver computes its mass and divides by it, so conditional normalization and both moment integrals are handled together. This also works at other valid given values of Y after updating the parameter and recomputing the slice.

2.8 STA 431 Problem 8: four tasks from one joint table

Quick load: group `HW5 Q5 TO Q8`, then `Q8 JOINT MOMENTS`. Choose `MOMENTS`; keep this dataset for the conditional part.

The PMF is $(x + y)/36$ on $x, y \in \{1, 2, 3\}$.

For both complete marginal tables, choose `MARG/CONDITIONAL > MARGINAL TABLES`.

Load once. Select `DISCRETE + JOINT > FORMULA/GRID`; enter `{1,2,3}` for each coordinate list and `X+Y` for the weight formula. The app generates nine rows and reports total 36 and normalizer 1/36.

Parts (a)–(c). Choose `MOMENTS` and advance through the X, Y, and product pages:

$$\boxed{\mathbb{E}X = \mathbb{E}Y = \frac{13}{6}, \qquad \mathbb{E}XY = \frac{14}{3}, \qquad \mathrm{Cov}(X, Y) = -\frac{1}{36}.}$$

The additional marginal variances are both 23/36, and correlation is $-1/23$; these are useful checks but are not requested by this question.

Part (d). Return to `TABLE TOOLS`, choose `MARG/CONDITIONAL > Y GIVEN X`, and enter `1`. The conditional-distribution page gives

y	1	2	3
$P(Y = y \mid X = 1)$	2/9	3/9	4/9

and the next page gives $\boxed{\mathbb{E}(Y \mid X = 1) = 20/9}$. These conditional weights sum to one.

Write the compact derivation. The marginals are

$$p_X(x) = \frac{x + 2}{12}, \quad p_Y(y) = \frac{y + 2}{12} \qquad (x, y = 1, 2, 3).$$

Thus $\mathbb{E}X = (3 + 8 + 15)/12 = 13/6$, with the same result for Y. Since $\sum_{k=1}^{3} k = 6$ and $\sum_{k=1}^{3} k^2 = 14$,

$$\mathbb{E}XY = \frac{14(6) + 6(14)}{36} = \frac{14}{3}, \qquad \mathrm{Cov}(X, Y) = \frac{14}{3} - \left(\frac{13}{6}\right)^2 = -\frac{1}{36}.$$

Finally $p_X(1) = 1/4$, so

$$p_{Y|X}(y \mid 1) = \frac{(1 + y)/36}{1/4} = \frac{1 + y}{9}, \quad y = 1, 2, 3,$$

with zero mass elsewhere, and $\mathbb{E}(Y \mid X = 1) = (2 + 6 + 12)/9 = 20/9$.

Why the app helps. A single formula-generated table serves all four parts. It computes both marginal moments, the product moment, covariance, the full requested conditional PMF, and its mean without re-entering the model.

2.9 Optional STA 631 Problem 8: zero covariance without independence

Quick load: group `OPTIONAL 631`, then `HW5 Q8 ZERO COV`. Choose moments, functions, or interval probability as described below.

This problem uses $f_X(x) = 1 - |x|$ on $(-1, 1)$ and $Y = X^2$. It is also a useful counterexample for the STA 431 independence topic.

App calculations. Select `DENSITY + INTEGRAL > 1D PDF OR SLICE`; enter `1-abs(X)`, fixed parameter `0`, and limits `(-)1`, `1`. The mass is 1. `MEAN+VARIANCE` gives $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1/6$. In `FUNCTION g(X)`, enter `X^3` to obtain $\mathbb{E}X^3 = 0$.

Part (a): the proof. Symmetry makes the first and third raw moments zero. Because $Y = X^2$,

$$\mathrm{Cov}(X, Y) = \mathbb{E}X^3 - \mathbb{E}X\mathbb{E}X^2 = 0 - 0 \cdot \frac{1}{6} = 0.$$

The symmetry argument supplies exact zeros; numerical output checks the moment calculation.

Part (b): build explicit events. Let

$$A = \{-1/2 < X < 1/2\}, \qquad B = \{0 \le Y < 1/4\}.$$

Because $Y = X^2$, the events coincide up to zero-probability endpoints. Use `INTERVAL PROB` with lower `(-)1/2` and upper `1/2`; the app gives

$$P(A) = \int_{-1/2}^{1/2} (1 - |x|)\,\mathrm{d}x = \frac{3}{4}.$$

Thus $P(B) = P(A \cap B) = 3/4$, but

$$\boxed{P(A \cap B) = \frac{3}{4} \neq \frac{9}{16} = P(A)P(B).}$$

The variables are dependent despite zero covariance.

Why the app helps. It evaluates the moment and interval integrals. The dependence proof comes from relating the two events through $Y = X^2$. Use the supplied one-variable density: the pair (X, X^2) lies on a curve and does not have an ordinary two-dimensional area density for the double-integral solver.

3 Improvements made while walking through the homework

The walkthrough was also a usability test. The following changes were implemented in the calculator package, rather than left as suggestions.

Friction found	Change made	Concrete benefit
Repeated model entry	`LOAD + SOLVE` provides input presets for all sixteen questions, using the same general calculators.	HW5 Q6 needs eleven numeric entries manually and zero through quick load; both produce variance 56.
Individual marginal queries	`MARGINAL TABLES` aggregates and prints both complete finite marginal distributions.	HW4 Q1's five marginal values appear in one command instead of five separate queries.
An unhelpful independence-failure message	The app now displays a violating pair, its joint probability, and its marginal product.	The displayed counterexample gives a concrete starting point for the exact written non-independence proof.
No direct continuous marginal/conditional workflow	The new `JOINT SLICE` tool keeps the original variable names, accounts for a known global multiplier, and reports marginal values and conditional results.	HW4 Q2 and the reimbursement problem can be checked without manually rewriting every conditional density as a one-variable model.
Repeated joint-model entry at a new conditioning value	`NEW GIVEN VALUE` reuses the shape and multiplier, then recomputes the slice with the bounds you enter.	HW5 Q7 can be checked at $y = 1/2$ and another y without retyping $X + Y$.
Dummy-variable output on a one-variable grid	A single y coordinate equal to zero is recognized as a univariate model.	HW5 Q1 no longer shows irrelevant Y moments and undefined-correlation pages.
A stored-function tokenization defect in an optional preset	The triangular density preset stores $1 - \sqrt{X^2}$ using executable calculator tokens.	The optional zero-covariance question loads correctly; this equals $1 - \lvert X \rvert$ for real X.

Where the app saves the most work

Finite-table arithmetic is highly automatable: normalization, all marginal masses, conditional probabilities, first/second moments, covariance, and transformed moments come from one loaded model. Given-moment questions such as HW5 Q5–Q6 are also ideal: the app expands the covariance terms and preserves their signs. The exact MGF templates handle the two assigned MGF models with minimal entry.

For joint continuous problems, the app removes most numerical integration and conditional-normalization arithmetic after the support and region have been selected. It still needs you to identify whether the region is the full support, an event, or a conditional slice. The homework's symbolic marginal formulas and independence proofs are short enough to write from the displayed setups; a numerical value cannot replace those functions or logical arguments.

Useful next improvements

Useful next additions would be a probability/complement shortcut for the device question and an optional support sketch. The current tools provide numerical results and analytical templates; symbolic density formulas and proofs remain in the written solution.